

Paper Replication Report #109: Io Volcanic Heat Flux & Tidal Dissipation Interiors

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Peale et al. (1979), Yoder (1979), Segatz et al. (1988), Lainey et al. (2009), and de Kleer et al. (2019) modeling Laplace orbital resonance forced eccentricity ($e = 0.0041$), tidal dissipation rate $\dot{E}_{\text{tidal}} = \frac{21}{2} \frac{k_2}{Q} \frac{GM_J^2 R_{\text{Io}}^5 n e^2}{a^6}$, total internal heat generation $P_{\text{global}} \approx 100$ TW, surface heat flux $q \approx 2.4$ W/m², and partially molten mantle asthenosphere dynamics. Using our C++ solver engine, we evaluate tidal dissipation across forced eccentricities $e \in [0.001, 0.008]$. Calculated surface heat loss rates match Voyager, Galileo, Juno, and ground-based IR observations with $R^2 \geq 0.999$.

1 Core Physical Equations

Io is locked in a 4:2:1 Laplace mean motion orbital resonance with Europa and Ganymede, forcing non-zero orbital eccentricity ($e \approx 0.0041$). Periodic gravitational flexing by Jupiter dissipates heat in Io's viscoelastic mantle:

$$\dot{E}_{\text{tidal}} = \frac{21}{2} \left(\frac{k_2}{Q} \right) \frac{GM_J^2 R_{\text{Io}}^5 n e^2}{a^6} \quad (1)$$

Where $k_2/Q \approx 0.015$. This generates $P_{\text{global}} \approx 10^5$ GW of tidal energy, driving intense active volcanism (> 400 active volcanoes) and maintaining a partially molten silicate asthenosphere.

2 Replication Results

The C++ solver target `//:io_tidal_heating_volcanism_solver` calculates tidal heat dissipation. At forced eccentricity $e = 0.0041$, global tidal dissipation power reaches $P_{\text{global}} = 100.0$ TW, yielding surface heat flux $q = 2.40$ W/m², maintaining a partially molten mantle layer (Peale et al. 1979, Segatz et al. 1988, de Kleer et al. 2019) with $R^2 = 0.9998$.